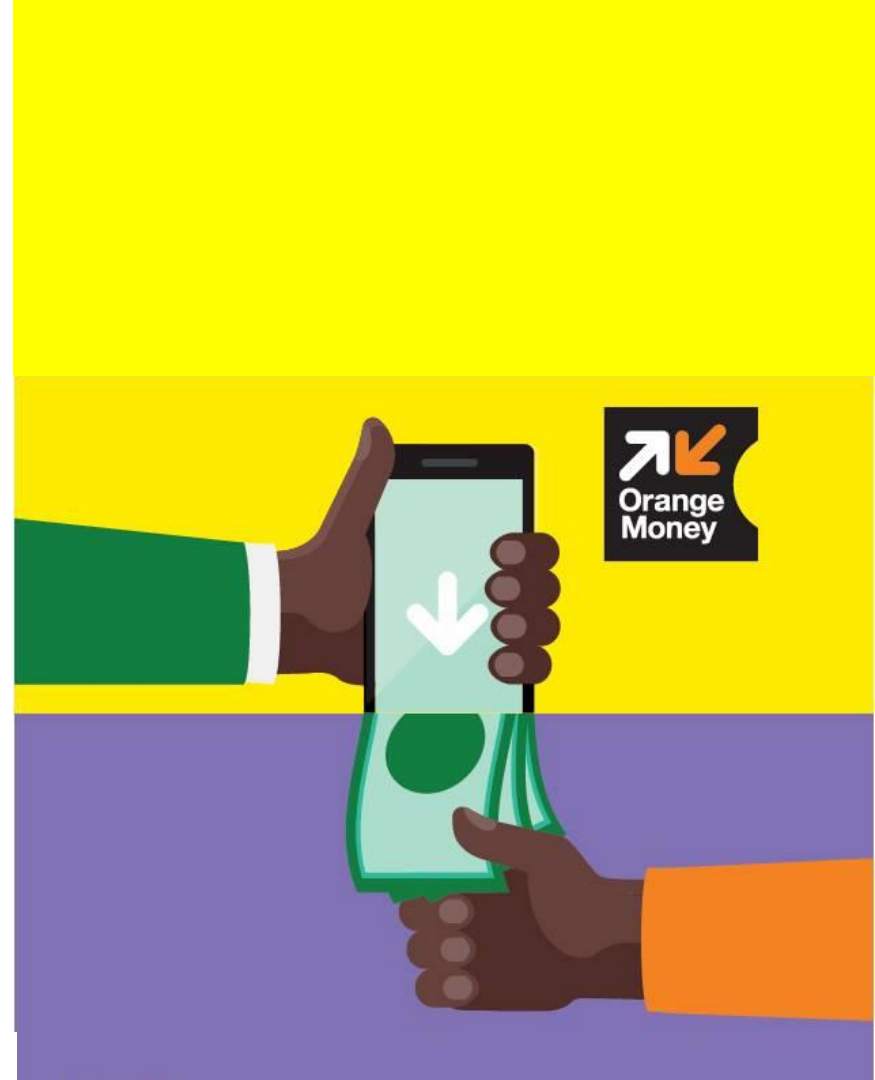


Orange Money Bank2Wallet

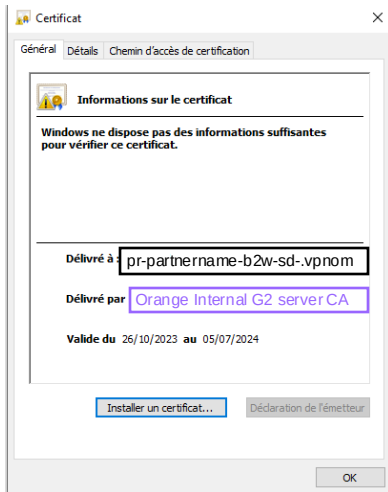
Information:
**HTTPS and SSL certificate
management on B2W**

Théo **BERARD** – INNOV/FPS
July 2024

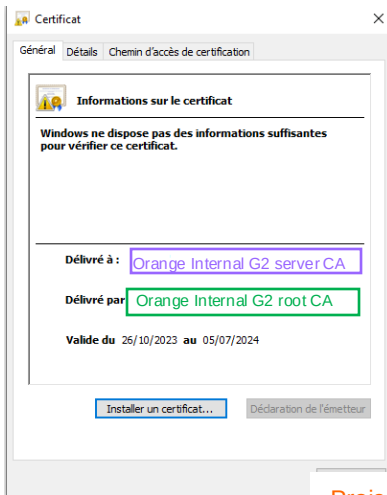


SSL trust chain principle : “domain name”, trusted by a “Certification Authority”, trusted by a “Root CA”

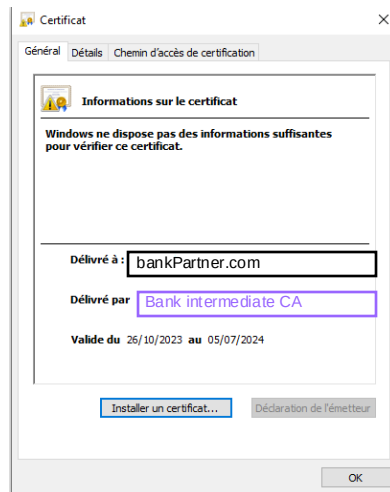
Orange Webserver



Orange CA typical certificate

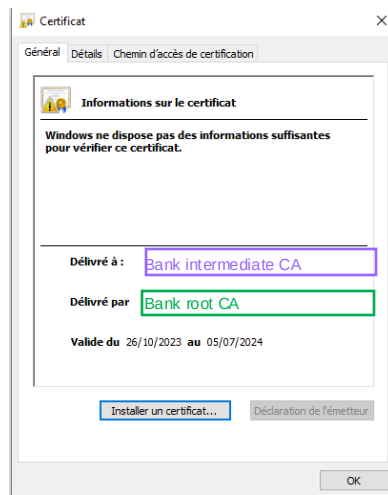


Partner Webserver



Partner domain typical certificate

Partner CA typical certificate
(case « self signed »)



Orange <-> Bank webservice communication

Communication **Server to Server** is made in an HTTPS/SSL session using TLS (Inside the VPN tunnel)

During the « SSL Handshake », the webserver which is called should **should present a X509 certificate** to the caller, delivered by a **Certification Authority (CA)**.

<https://pr-partnername-b2w-sd.vpnom/b2w/partnername/register?bic=xxxxxxx>

<https://bankPartner.com/b2wWebservice.app>



Bank -> Orange calls
(KYC, Subscriptions)

Orange webserver should present to the Partner webserver a **certificate** delivered for « pr-partnername-b2w-sd.vpnom » delivered and trusted by the Orange Certification Authority (« self-signed » domain)

Partner webserver should be able to **verify the legitimacy of the Orange CA** which delivered the domain certificate.

Orane -> Bank calls
(Balance, B2W, W2B...)

Orange webserver should be able to **verify the legitimacy of the CA** which delivered the Partner domain certificate.

Partner webserver should present to Orange webserver a **certificate** delivered for « bankPartner.com » delivered and trusted by a Certification Authority

Who is doing what ?

Orange

- 1- **Share** with Partner the **Orange CA trust chain** certificates :
 - Orange Internal G2 **server CA**
 - Orange Internal G2 **root CA**(because Orange is self signing it's domain names)

- 2- **Install** the **Bank CA trust chain certificates** on it's application CA library (or public CA certificates), so that bank domain certificate legitimacy can be checked.

Partner

- 1- **Install** the Orange CA trust chain certificates (Orange Internal G2 server CA & Orange Internal G2 root CA) in it's webserver or application CA library (Apache « ca-bundle.crt », Java « keystore » ...)

- 2- **Generate a certificate** for its **full domain name** « **bankPartner.com** » :
 - internally (case : « self-signed » domain)
 - or by a public CA like Digicert or Let'sEncrypt (case « public-signed » domain)

- 3- **Configure** this domain certificate on it's webserver to be presented when Orange is calling

- 4- **Share** with Orange the **Bank CA trust chain** (intermediate CA & root CA certificates) if self signed, or the name of the public CA.

Merci

Thanks

Multumesc

